Reset Early, Reset Often, Saturate the SAT Core

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Abstract

Connection tableau provers usually try to close one tableau before moving on to another branch of the search. We introduce SATResetCoP, a variant of SATCoP that instead treats the accumulated SAT abstraction as the main search object: it repeatedly resets the active tableau and uses the new ground clauses produced by these attempts to drive the SAT abstraction toward contradiction.

Fredrik: What is the difference between this and resolution based theorem proving? It seems we are moving SATCoP more towards the saturation domain. Especially relevant for a Vampire workshop.

We compare this against implementations of leanCoP and SATCoP in the same backend framework. SATResetCoP solves 2971 problems under limits of 120 seconds and 1 million steps, an improvement of 54% over our implementation of leanCoP and 18% over our improved implementation of SATCoP.

Fredrik: I would stick to percentage improvements on benchmarks (TPTP,M2k,MPTP) and save seconds and steps to when we can define what steps are

These results suggest that SAT saturation is useful as tableau search guidance.

Fredrik: This was the claim of SATCoP, what are the claims we add to that?

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Fredrik: Are ToCs standard in the Vampire workshop? If not I would remove.

1 Introduction

Connection tableau provers are attractive because the core proof search is small and goal-directed.

Fredrik: proof search is small?

This is especially visible in leanCoP, where a competitive first-order prover can be implemented with very little machinery[11, 10]. The cost is that the prover has to make many local control decisions: which clause to start from, which connection to try next, when to backtrack, and when to restart.

Fredrik: There is a lot of machinery behind Prolog which leanCoP takes advantage of. This hardcoded SLD-resolution style search is perhaps what we want to aleviate with the reseting

These choices matter because connection proof search is not confluent. A locally valid connection can bind variables in a way that makes another branch impossible to close later. leanCoP's `cut` option shows that even a crude restriction on backtracking can make a large practical difference, although it loses completeness unless it is used inside a complete schedule[9].

SATCoP changes the objective slightly. Instead of using the tableau only as a proof object that must eventually close, it also uses the tableau as a source of ground clauses for an incremental SAT solver[16]. Once the accumulated ground clauses are propositionally unsatisfiable, the prover can stop even if the current first-order tableau has not closed. This suggests a different view of tableau control: a good run is not only one that closes the tableau quickly, but one that generates useful ground instances quickly.

Fredrik: Good paragraph. We may want to specify that our prover works in positive representation, so that we are making the sat solver find a valid core rather than an unsat core (polarities flipped). Though this may just add confusion.

This paper explores that view by resetting the tableau aggressively. If the current bounded search stops adding new ground information to the SAT solver, we abandon it and start another search. This is incomplete as a standalone proof procedure, but useful when the goal is to saturate the SAT core fast.

Fredrik: UNSAT core?

The rest of the paper describes the resulting prover, SATResetCoP, and compares it against leanCoP-style and SATCoP-style baselines on TPTP problems.

2 Preliminaries

2.1 Connection Calculus and Tableau Solving Methods

Connection tableaux and model elimination give a goal-directed alternative to the saturation style used by provers such as Vampire[17]. Instead of repeatedly deriving clauses from a global clause set until a contradiction is found or the search space saturates[1, 20], connection calculi try to close a tableau by connecting goal literals to complementary literals, either on the current path or in a fresh copy of an input clause[7, 12].

The standard clausal connection tableau has three proof-building inferences[7, 13].

Start Choose an input clause and make its literals open goals.

Extension Open a branch on a clause containing a complementary literal, and add the remaining literals as new subgoals. This might require a substitution of variables with terms.

Reduction Closes the current goal against a complementary literal on the branch. Like the previous step, this might require adding a substitution.

The calculus is sound and complete with backtracking and iterative deepening, but it is not confluent: a locally valid connection can bind variables in a way that prevents later branches from closing[15].

Fredrik: I would cite [7] for proof confluence. Sections 2.3 and 2.4 of that goes over this in detail.

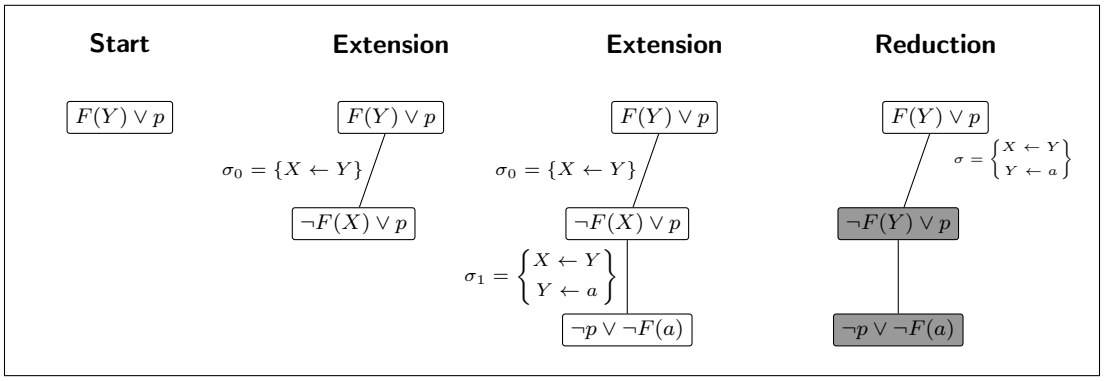


Figure 1: The three proof-building steps of a clausal connection tableau, taken from the tableau closure of SYN052-1.p[14]. The two Extension steps add an extra clause to the tableau; the Reduction step closes the nodes against a complementary literal already on the path. A full solution can, in several different strategies, be found in Figure 6.

Many different connection tableau provers have been developed over the years. leanCoP adapted this style of connection tableau proving to Prolog and became the basis of a large family of provers for classical and non-classical logics[11, 10].

leanCoP comes with several optional predicates. Most importantly, `cut` commits a Prolog cut `(!)` after every successful Reduction and Extension. This removes many alternative local continuations and makes the base prover incomplete, but it also converges to a solution considerably faster[9]. Additionally, leanCoP iteratively increases the maximum tableau depth when the current options are saturated.

To keep the prover complete, the `comp(I)` option removes the maximum height restrictions and the `cut` when starting a tree that would have had maximum height `I`.

Later systems can be read as attempts to make this local control less crude. meanCoP separates cut on reductions from cut on extensions and introduces exclusive cuts, which preserve more alternatives than the inclusive cuts used by leanCoP[4]. hopCoP takes the opposite route: instead of deliberately throwing away alternatives, it learns constraints explaining why a search state is stuck, thereby reducing repeated backtracking while preserving completeness at a fixed iterative-deepening level[15].

Functional connection provers, mlCoP, and monteCoP reorder proof steps using learned information or Monte Carlo Tree Search[5]. rlCoP and plCoP make this more explicit by treating proof search as a learning-guided control process over leanCoP-style actions[6, 23, 24]. The common theme is the one we use below: connection-tableau proving is naturally viewed as a sequence of small control decisions over proof actions, cuts, backtracking, and restarts.

2.2 SATCoP and Methods that use SAT Saturation in a Tableau

SAT and SMT solvers have long been used to support first-order reasoning[16, 19]. For a connection prover, the useful observation is simple: every clause that appears in a tableau is an instance of an input clause. If we ground these tableau clauses and collect them in a propositional SAT solver, then a SAT refutation of the collected ground clauses is a sound refutation of the original first-order problem.

Rawson and Reger implement this idea in SATCoP, a clausal connection-tableau prover based on leanCoP[16].

During backtracking search, clauses present in the tableau are heuristically grounded and added to an incremental SAT solver. If the SAT solver reports that the accumulated ground clauses are unsatisfiable, proof search can stop even if the first-order tableau itself has not been closed[16]. The remaining tableau variables must be mapped to ground terms before a clause is sent to SAT. Their implementation follows a Schulz-style heuristic: use the most frequent constant in the conjecture, and fall back to a fresh constant if no conjecture constant is available[16, 19, § 4.2].

Later modifications to SATCoP gave it control of the tableau using results from the SAT solver. In a later note included in the same paper, Rawson and Reger also describe a literal-selection refinement: prefer extension choices whose selected literal is true in the current SAT model[16, § 9.1].

The strategy used in this paper, SATResetCoP, pushes the ideas from SATCoP further by resetting trees as soon as the SAT model stops growing.

3 Research Vehicle

3.1 SATResetCoP Specification

We propose a new tableau prover based on SATCoP, but guided by the goal of saturating the SAT solver rather than closing the current tableau.

Successive experimentation led to a simple strategy for achieving this: aggressively reset the tree when the SAT solver stops receiving new ground clauses. This adds a broader set of ground clauses to the solver in less time, allowing the SAT solver to reach unsatisfiability earlier.

SATResetCoP uses many of the properties of SATCoP described in the original paper and in its "Future Research" section[16, § 9.1], but changes the tableau guidance to prioritise aggressive resets.

Just like PySATCoP, this approach is sound. While it is less complete than PySATCoP and other leanCoP variants, the speed at which it proves unsatisfiability can increase the number of problems solved within fixed time and step limits. It might be useful to incorporate SATResetCoP into the initial steps of a more complete schedule.

A concrete example of how PyLeanCoP, PySATCoP, and SATResetCoP handle SYN052-1.p can be found in Figure 6.

Fredrik: I don't believe PySATCoP and PyLeanCoP have been introduced. I think in general there should be a reshuffling of the order in which things are introduced and detailed in sections 3 and 4.

3.2 Implementation Details

Like SATCoP, the implementation keeps a persistent SAT solver alongside the active tableau search. The SAT state is not recreated during ordinary proof search. Ground clauses accumulated in earlier tableau attempts remain available in later attempts, and new clauses are added incrementally to the same solver instance.

The SAT state is implemented using CaDiCaL[2]. A new virtual operation, Reset, clears the tableau, the current substitution, and the proof sequence, and starts a new tableau attempt with a specified depth bound. It does not clear the persistent SAT solver: every reset starts from an empty tree, while the SAT solver keeps the ground information produced by earlier attempts. An example of how this operation works can be seen in Figure 2.

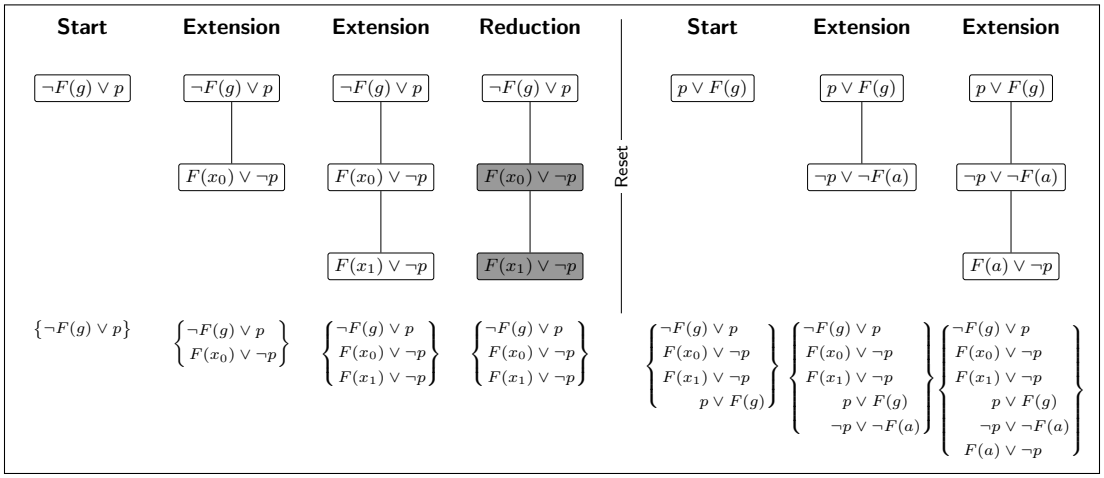


Figure 2: Example of SATResetCoP on SYN052-1.p. SATCoP and previous provers would backtrack after closing the branch with $F(x_1) \vee \neg p$; instead, SATResetCoP resets into a new tree. This grows the UNSAT core shown at the bottom faster than in SATCoP, allowing the SAT solver to reach unsatisfiability earlier. Substitutions are omitted for simplicity.

The grounded clauses used by PySATCoP and SATResetCoP come from clauses that have appeared in the tableau, after applying the tableau substitution accumulated so far. Start clauses are added to the SAT state when the run is initialised. After successful Extension and Reduction steps, the implementation grounds the clauses currently present in the tableau and records the genuinely new ones, which are later flushed to the SAT solver.

The grounding scheme follows the soundness argument of Section 2.2: the SAT solver receives only ground instances of clauses generated by the tableau. Ground terms are kept as they are, and variables assigned by the current tableau substitution are replaced by their assigned terms. Only tableau variables that remain unassigned need to be replaced by a ground term. Rawson and Reger follow the technique by Schulz[19] and use the most frequent constant in the conjecture, with a fresh constant as a fallback when the conjecture contains no constants[16, § 4.2]. SATResetCoP always uses this fallback-style choice, mapping every remaining tableau variable to the same fresh artificial constant. This gives a grounding scheme that is simpler than SATCoP, while keeping the same soundness condition.

The reset policy is the main implementation difference between SATResetCoP and PySATCoP. At a dead end, the prover first asks whether the current tableau attempt produced any new ground clauses that have not yet been added to the SAT solver. If it did, those clauses are flushed

```
def solve_tableau(start_clauses):
  solver := incremental_sat_solver()
  solver.add(ground(start_clauses))
  pending := empty_set

  max_depth := 1
  tableau := empty_tableau(max_depth)

  while solve_sat(solver) != UNSAT:
    model := current_model(solver)
    actions := legal_actions(tableau)

    if any Start, Extension, Reduction in actions:
      model_rank(a) :=
        2 if a is Start
          0 if selected_literal(a) is true in model
        1 if selected_literal(a) is unknown in model
        3 if selected_literal(a) is false in model

      next_action = argmin(actions, key = model_rank)
      tableau_step(next_action)

      if next_action is Reduction or Extension:
        pending.add(ground(tableau))

    else:
      if pending is not empty:
        solver.add_all(pending)
        pending := empty_set
        tableau := empty_tableau(max_depth)
      else:
        max_depth := max_depth + 1
        tableau := empty_tableau(max_depth)
```

Listing 1: Pseudocode of the SATResetCoP control policy.

to the SAT solver and the prover resets at the same depth. This gives the next tableau attempt the same height bound, but a larger SAT state and a fresh randomised ordering of choices. If the attempt did not produce any new ground clauses, the prover treats the current depth as exhausted and resets at the next depth instead of spending more steps on local backtracking. The process stops as soon as the accumulated ground clauses become propositionally unsatisfiable.

SATResetCoP randomises the Start and Extension steps in the same way as SATCoP. When starting a new tree, the implementation shuffles the possible starts and the literals inside each start clause. When several extensions are possible from the leftmost open node, it shuffles the extension choices and the literal order inside the copied extension clause. Reductions are not randomised in the current implementation. The random generator is seeded, so the experiments remain reproducible.

Pseudocode for this implementation can be found in Listing 1.

3.3 Python Implementation

As a testbed for our experiments we use the Connections Prover by Rømming et. al[18], which provides an implementation of leanCoP-style provers. The base prover can be found in

https://github.com/fredrrom/connections.

Fredrik: .." we use the Connections library by Rømming et. al[18], which provides primitives for the implementation of connection tableau provers.."

The connections prover makes it easy to implement different strategies for guiding the tableau proof. This keeps the SATResetCoP implementation simple, readable, and easy to evaluate.

The source code for this implementation is available at https://github.com/mfixman/resetsatcop, and is released under the GNU Lesser General Public Licence[3].

3.4 TPTP Benchmarking

Along with our implementation of SATResetCoP, PyCoP allows us to reimplement PyLeanCoP and PySATCoP in the same backend. This provides a fairer baseline for comparison.

Fredrik: "Connections allows us to reimplement stepwise equivalent veersions of LeanCoP and SATCoP in the same backend."

We run SATResetCoP, our two connection-tableau baselines, and a simple strategy that chooses the first possible action on 7815 CNF problems and 8779 FOF problems from the Thousands of Problems for Theorem Provers (TPTP) library[21]. These problems cover all 50 TPTP categories containing at least one CNF or FOF problem.

The library provides a varied problem set that is a standard benchmark for first-order provers. It also provides declared statuses for these problems, which are a useful sanity check for false unsatisfiability results.

4 Experimental Evaluation

4.1 Experiment Design

The connections prover includes PyCoP, a simple control policy that chooses the first legal non-backtrack action at each point and resets the tree with an increased maximum height when the current tree saturates. This serves as a baseline and sanity check.

Fredrik: I would avoid referencing pyCoP and focus on the fact that we have stepwise equivalent versions of leanCoP and SATCoP in the same tableau constructor backend

As a second control policy we use our implementation of leanCoP[9], PyLeanCoP. This is a Python version of the leanCoP Prolog implementation, with the two recommended predicates used in the first step of the schedule by Otten[9, § 5.5].

`cut` Choose one extension or reduction and commit to it, similarly to Prolog's `!` operator.

`comp(7)` The prover increases the depth bound by 1 whenever the current tree saturates, until reaching maximum height 7. Afterwards, the height limit is removed.

These simple policies tend to be very effective in tableau provers[10].

The benchmark set contains 7815 CNF problems and 8779 FOF problems spread across 50 categories[21]. Each problem-model pair is run for at most 120 seconds and 1 million steps, on a single core of the University of Cambridge CSD3 Cluster with [add CPU information] and 3370 MiB RAM per core[22].

Model	Explanation	Cut	UNSAT Proof	SAT Guidance	Reset on Backtrack
PyCoP PyLeanCoP	Simple control policy leanCoP with initial set of predicates.	✓			
PySATCoP$_0$	Based on the initial implementation of SATCoP[16].	✓	✓		
PySATCoP	Improvements on SATCoP[16, § 9.1].	✓	✓	✓	
SATResetCoP	Novel model, specified in Section 3.1.	✓	✓	✓	✓

Table 1: The four control models and the test model used in this experiment.

4.2 Results

4.2.1 Individual Results

SATResetCoP considerably increases the number of problems solved by both PySATCoP$_0$ and PySATCoP under the 120-second time limit, as shown in Table 2. The main driver is that the SAT solver reaches unsatisfiability in less time and fewer steps than in the previous models, as seen in Figures 3a and 3b. The 1 000 000 step limit is rarely reached, and most unsolved problems hit the 120 second limit instead.

The improvement is visible in both problem formats. SATResetCoP solves 462 more problems than PySATCoP, with 206 additional CNF problems and 256 additional FOF problems. Compared with PySATCoP$_0$, it solves 1100 more problems, an increase of 58.8%, and compared with PyLeanCoP it solves 1048 more problems, an increase of 54.5%.

	CNF		FOF		Total	
Model	Number	%	Number	%	Number	%
PyCoP	208	2.7	587	6.7	795	4.8
PyLeanCoP	723	9.3	1200	13.7	1923	11.6
PySATCoP$_0$	804	10.3	1067	12.2	1871	11.3
PySATCoP	1067	13.7	1442	16.4	2509	15.1
SATResetCoP	1273	16.3	1698	19.3	2971	17.9

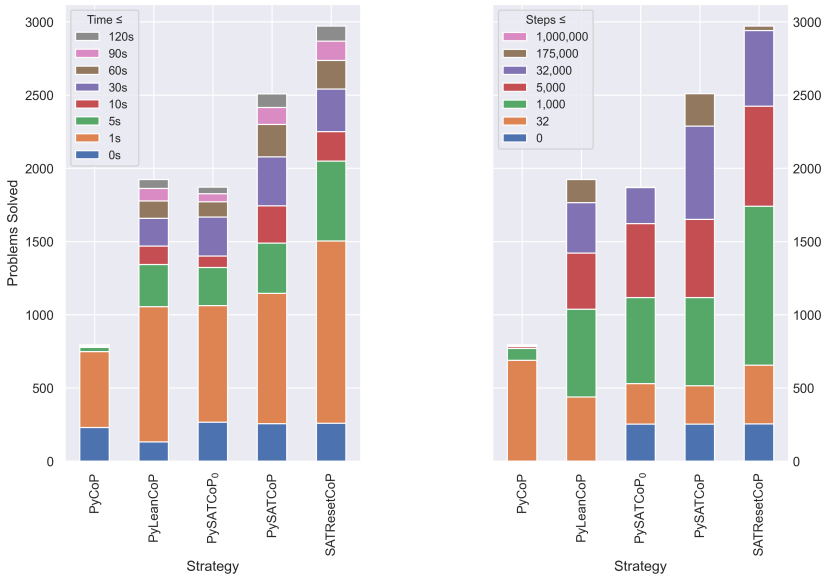
Table 2: Number and percentage of CNF and FOF problems solved within the time limit by each model. Each added strategy component improves results on both CNF and FOF problems.

4.2.2 Comparative Results

The SAT guidance in SATResetCoP allows it to solve several problems left unsolved by models without the same SAT saturation mechanism, and even by PySATCoP$_0$ and PySATCoP, which use less aggressive guidance. A comparison of newly solved and newly unsolved problems can be seen in Table 3.

Figure 4 shows the number of problems solved by one model but not by another.

The improvement between SATResetCoP and the best previous experimental model, our PySATCoP implementation of SATCoP, is especially visible in TPTP categories where propositional contradictions are reached quickly in the accumulated ground abstraction, such as SYN,



(a) Problems solved, stacked by time in seconds.

(b) Problems solved, stacked by number of steps. SAT models can solve a problem in 0 steps if the initial grounding is already SAT-unsatisfiable.

Figure 3: Results of running the models on all problems under limits of 120 seconds and 1 million steps.

Model	Newly Solved	Newly Unsolved	Total Solved
PyCoP			795
PyLeanCoP	1246	118	1923
PySATCoP$_0$	307	379	1871
PySATCoP	402	23	2509
SATResetCoP	478	136	2971

Table 3: Newly solved, newly unsolved, and total solved problems for each model.

SWC, and SWV. As expected, this strategy also solves problems in fewer steps, as can be seen in Figure 5.

Figure 6 shows the solution to the TPTP problem SYN052-1[14] in three different strategies. This is an example of the kind of problem on which SATResetCoP performs best, since it can be solved with a relatively small UNSAT core.

Model	SYN	SWC	SWV	GEO	NUM	Rest	Total
PyCoP	231	30	129	28	9	368	795
PyLeanCoP	486	14	163	216	86	958	1923
PySATCoP$_0$	459	2	161	177	90	982	1871
PySATCoP	549	11	173	260	151	1365	2509
SATResetCoP	672	64	215	293	179	1548	2971

Table 4: Number of problems solved, per category, for each model.

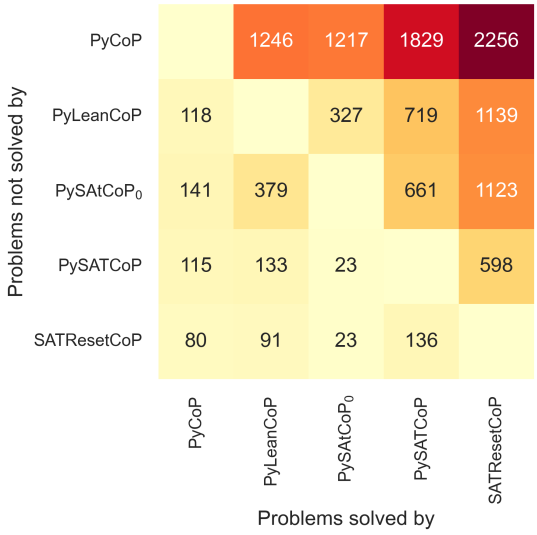


Figure 4: Pairwise comparison of model results: the cell in column y and row x shows the number of problems solved by model y, but not by model x. For example, SATResetCoP solved 1123 TPTP problems that PySATCoP$_0$ did not.

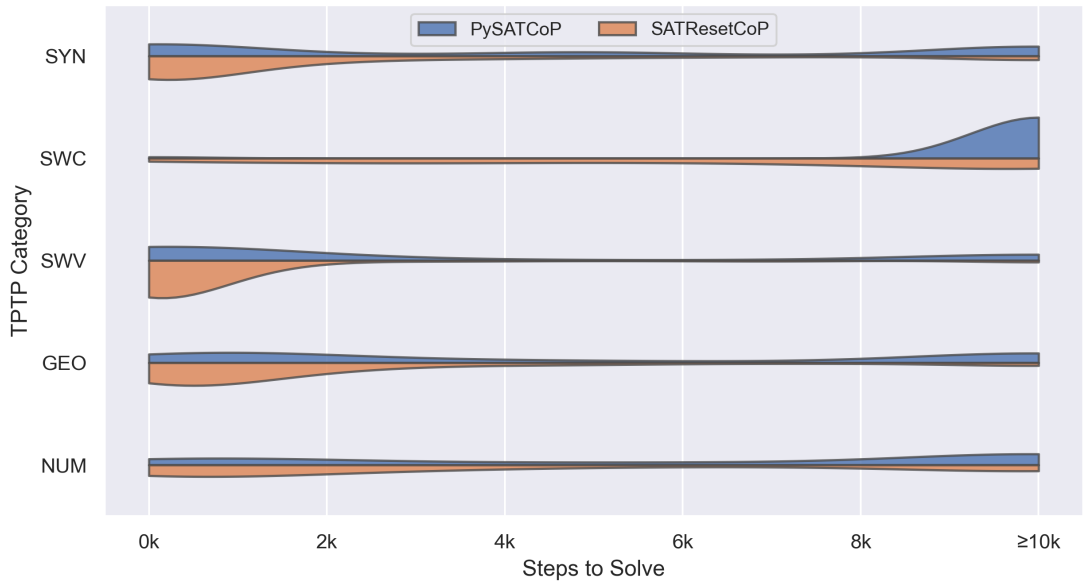
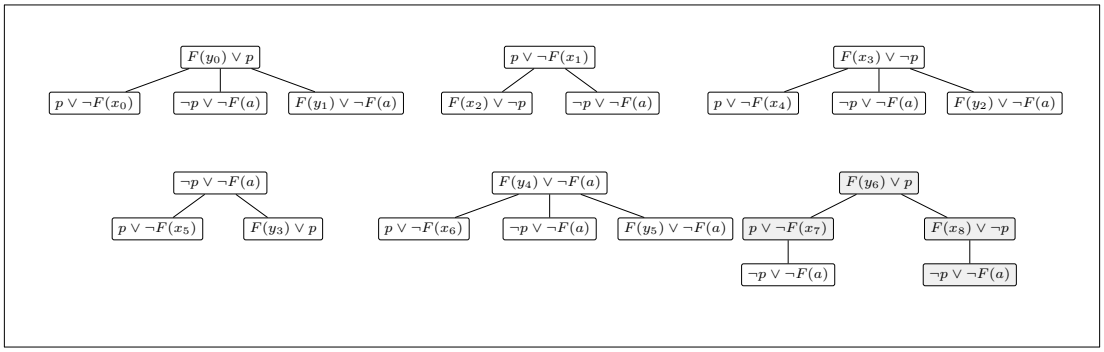


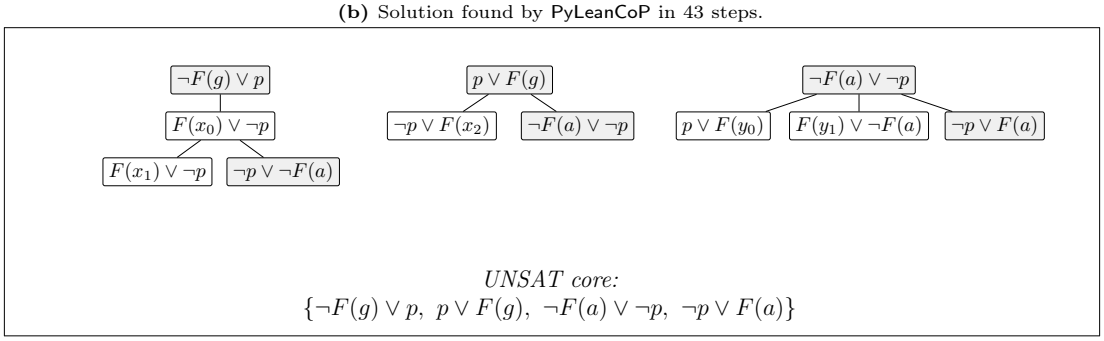
Figure 5: Density of problems solved at each step count by PySATCoP and SATResetCoP on the TPTP categories SYN, SWC, SWV, GEO, and NUM. The area of each violinplot is the same; besides solving more problems in each category, SATResetCoP solves many problems in fewer steps. Unsolved problems are collected in the $\geq 10k$ category on the right.

$$p \vee \neg F(X) \qquad F(X) \vee \neg p$$
$$F(Y) \vee p \qquad \neg p \vee \neg F(a) \qquad F(Y) \vee \neg F(a)$$

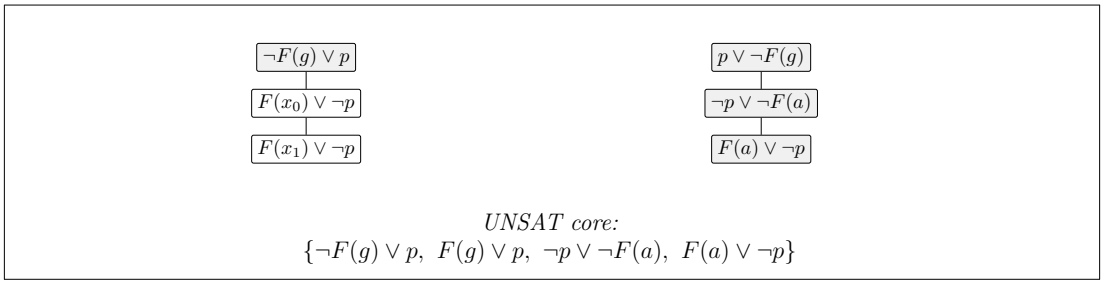
(a) Problem SYN052-1.p[14], adapted from https://tptp.org/TPTP/Problems/SYN/SYN052-1.p.



(b) Solution found by PyLeanCoP in 43 steps.



(c) Solution found by PySATCoP in 34 steps.



(d) Solution found by SATResetCoP in 11 steps.

Figure 6: Unsatisfiability proofs for the TPTP problem SYN052-1[14] produced by three provers: PyLeanCoP, PySATCoP, and SATResetCoP. [Should this be in an appendix?]

	FOF		CNF	
Declared status	Unsolved	Solved	Unsolved	Solved
Satisfiable	350	0	1094	0
CounterSatisfiable	732	0	0	0
Unknown	94	0	745	0
Open	4	0	28	0
Unsatisfiable	100	72	4496	1433
ContradictoryAxioms	204	39	0	0
Theorem	4426	1823	0	0

Table 5: Number of problems solved and unsolved by SATResetCoP, grouped by TPTP declared status. No satisfiable or countersatisfiable problem was incorrectly proven to be unsatisfiable.

4.2.3 Soundness Verification: Problem Satisfiability

The TPTP library includes declared statuses for its problems[21]. While this does not demonstrate soundness, it provides a useful check that the strategy has not become unsound by mistake.

We make a second run of SATResetCoP with a longer 10-minute limit, whose results can be seen in Table 5. The results are consistent with what we would expect if the prover were sound, as it does not find unsatisfiability ("Solved") in any problem marked satisfiable or countersatisfiable. It also does not solve any unknown or open problems; while this limits the number of new discoveries, any such solutions would need to be checked closely to ensure that they were not caused by an incorrect prover.

5 Future Work

SATResetCoP still struggles with a significant number of problems, as seen in Table 3. It would be worth experimenting with adding this strategy to a schedule, following the examples of Otten and Mangla[9, 8], and possibly running a different strategy or solver if no solution is found after a fixed amount of time or number of steps.

6 Acknowledgements

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